

Self-Knowledge Guided Retrieval Augmentation for Large Language Models

Yile Wang¹, Peng Li^{*1,4}, Maosong Sun^{2,3}, Yang Liu^{*1,2,3,4}

¹Institute for AI Industry Research (AIR), Tsinghua University, Beijing, China

²Dept. of Comp. Sci. & Tech., Institute for AI, Tsinghua University, Beijing, China

³Beijing National Research Center for Information Science and Technology

⁴Shanghai Artificial Intelligence Laboratory, Shanghai, China

{wangyile, lipeng}@air.tsinghua.edu.cn, {sms, liuyang2011}@tsinghua.edu.cn

Abstract

Large language models (LLMs) have shown superior performance without task-specific fine-tuning. Despite the success, the knowledge stored in the parameters of LLMs could still be incomplete and difficult to update due to the computational costs. As complementary, retrieval-based methods can offer non-parametric world knowledge and improve the performance on tasks such as question answering. However, we find that the retrieved knowledge does not always help and even has a negative impact on original responses occasionally. To better make use of both internal knowledge and external world knowledge, we investigate eliciting the model’s ability to recognize what they know and do not know (which is also called “self-knowledge”) and propose Self-Knowledge guided Retrieval augmentation (SKR), a simple yet effective method which can let LLMs refer to the questions they have previously encountered and adaptively call for external resources when dealing with new questions. We evaluate SKR on multiple datasets and demonstrate that it outperforms chain-of-thought based and fully retrieval-based methods by using either InstructGPT or ChatGPT.

1 Introduction

Large language models (LLMs, Brown et al., 2020; Chowdhery et al., 2022; Ouyang et al., 2022) have achieved remarkable performance without much task-specific fine-tuning. However, the full-parametric knowledge stored in LLMs could still be incomplete and difficult to update due to the computational costs. Alternatively, retrieval-augmented methods (Guu et al., 2020; Lewis et al., 2020b; Borgeaud et al., 2022; Izacard et al., 2022; Shi et al., 2023) can utilize external resources such as Wikipedia and offer complementary non-parametric knowledge to enrich the contextualized

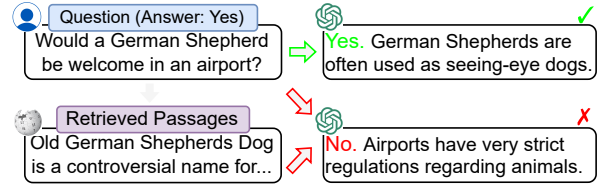


Figure 1: Comparison between two responses given by InstructGPT. The retrieved passages are relevant but not particularly helpful for solving the question, which influences the model’s judgment and leads to incorrect answers.

information, thus helping the model generate more reliable answers.

Retrieval augmentation has shown to be very effective for models such as BERT (Devlin et al., 2019), BART (Lewis et al., 2020a), and T5 (Raffel et al., 2020) in various tasks (Karpukhin et al., 2020; Khandelwal et al., 2020, 2021; Izacard and Grave, 2021; Wang et al., 2022; Guo et al., 2023). As LLMs become more and more “knowledgeable”, recent studies show that the benefit brought from retrieval augmentation is reducing (Mallen et al., 2022; Yoran et al., 2023). Moreover, we find that the retrieved passages could even negatively affect what LLMs originally know. As illustrated in Figure 1, the model can directly give reasonable answers “German Shepherds are often used as seeing-eye dogs”, however, it is distracted and gives incorrect ones by adding retrieved passages.

The above findings show that one should be more careful when applying the retrieval-based method since it is difficult to know in advance whether the retrieved results are better than what LLMs already captured. To this end, a key issue is to figure out what LLMs do well (e.g., they can answer correctly without assistance) and what they cannot do well (e.g., they answer incorrectly and external information can lead to improved results).

Unfortunately, LLMs themselves have a limited ability to recognize what they know and do not

* Corresponding authors.

know, which is also called “self-knowledge” (Yin et al., 2023). However, such an ability is crucial for generating truthful responses (Kadavath et al., 2022) and could be helpful for LLMs themselves to “decide when and when not to use tools” such as a retriever (Mialon et al., 2023).

In this paper, we investigate eliciting the self-knowledge of LLMs and propose a simple yet effective Self-Knowledge guided Retrieval augmentation (SKR) method to flexibly call the retriever for making better use of both internal and external knowledge. In particular, different from existing studies that evaluate the ability through specifically-designed metrics or datasets, we collect the self-knowledge of training questions by comparing the performance with or without retrieval augmentation. Then, we propose several strategies to detect the self-knowledge corresponding to a question by referring to the existing collected training questions, including using the LLMs themselves through prompting or explicitly training a small model. Finally, we leverage such elicited self-knowledge to better solve the question through adaptive retrieval augmentation.

We evaluate SKR on five datasets by using InstructGPT (text-davinci-003) and ChatGPT (gpt-3.5-turbo-0301). Experimental results show that SKR outperforms chain-of-thought based (Wei et al., 2022) and fully retrieval-based methods by 4.08%/2.91% (for InstructGPT) and 4.02%/4.20% (for ChatGPT), respectively.

2 Related Work

Retrieval-Augmented LLMs Recent studies show that retrieval-augmented methods can enhance the reasoning ability of LLMs (Trivedi et al., 2022; He et al., 2022; Yu et al., 2023; Shao et al., 2023; Jiang et al., 2023) and make the responses more credible and traceable (Xu et al., 2023b; Qian et al., 2023). For example, Trivedi et al. (2022) uses the chain-of-thought (Wei et al., 2022) reasoning steps as queries and uses the results to guide further reasoning and retrieval. He et al. (2022) uses an external natural language inference model to select the most supported reasoning path via retrieved evidence. Yu et al. (2023) propose using the retrieval feedback to refine the output of LLMs to be more reliable and accurate. Xu et al. (2023b) propose search-in-chain and make LLMs interact with retrievers to improve accuracy and credibility. These methods aim at integrating sufficient external knowledge for

a better reasoning process, while we propose to better utilize both the internal and external knowledge through eliciting the self-knowledge of LLMs.

Another line of work tries to teach LLMs to use external tools including retriever, calculator, other foundation models, etc. (Schick et al., 2023; Shen et al., 2023; Qin et al., 2023). These works focus more on leveraging the language understanding capabilities of LLMs to deploy suitable tools in different scenarios, while our work investigates the self-knowledge of LLMs and tries to integrate them with retrievers in a more flexible manner.

Self-Knowledge in LLMs “Self-knowledge” in LLMs is originally mentioned in Kadavath et al. (2022), which is used to measure the LLMs’ confidence in their own knowledge and reasoning. Such ability is further defined as “the ability to understand limitations on the unknowns” and evaluated by Yin et al. (2023), where they find a considerable gap exists between self-knowledge in models and humans. To explore the LLMs capabilities more extensively, unanswerable and more challenging datasets are also proposed (Rajpurkar et al., 2018; Srivastava et al., 2022; Suzgun et al., 2022). Our work is also related to detecting what LLMs know and do not know, while we do not design new evaluation metrics or challenging datasets to test the ability. By explicitly introducing the external resources, we detect the knowledge boundary of LLMs through the performance changes. Moreover, instead of evaluating each question independently, we propose several ways to elicit self-knowledge by referring to existing cases.

3 Method

Our method is depicted under the question-answering settings, which has been a popular way to interact with and assess LLMs. The overall pipeline is shown in Figure 2, which includes collecting, eliciting, and using self-knowledge of LLMs. We introduce each of them as follows.

3.1 Collecting Self-Knowledge of LLMs from Training Samples

Given a dataset $\mathcal{D}$ with training question-answer pairs $\{q_j, a_j\}_{j=1}^{|\mathcal{D}|}$, we can use the LLM $\mathcal{M}$ to generate the answers for each question q_i via few-shot in-context learning (Brown et al., 2020):

$$\hat{a}(\mathcal{M}, q_i) = \mathcal{M}(q_1 \circ a_1, \dots, q_d \circ a_d, q_i), \quad (1)$$

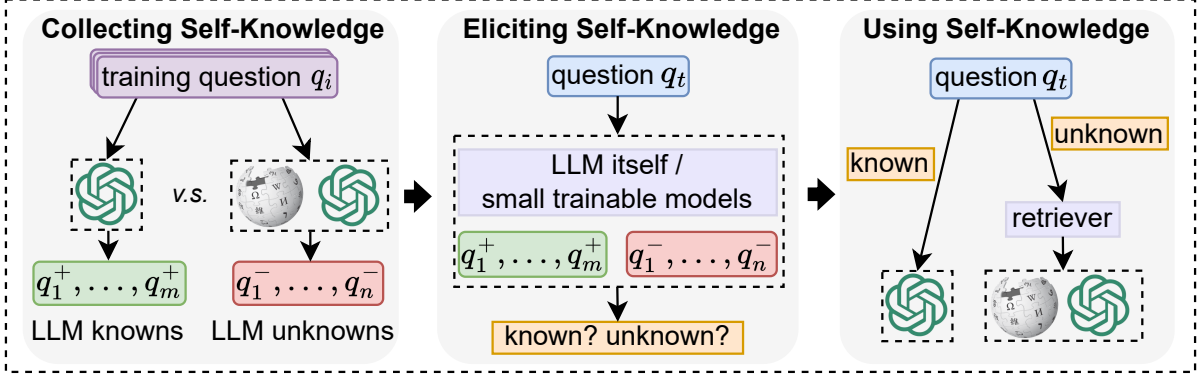


Figure 2: The overall pipeline of our SKR method. We first collect self-knowledge from training questions according to the performance with or without external information (§ 3.1). Then we use the LLMs themselves or explicit small trainable models to elicit self-knowledge of a question q_t by referring to the collected self-knowledge from training questions (§ 3.2). Finally, we use the self-knowledge to the new question and adaptively call a retriever (§ 3.3).

where $\circ$ denotes concatenation and $\{q_j \circ a_j\}_{j=1}^d$ are d demonstrations.

The above generated answers $\hat{a}(\mathcal{M}, q_i)$ reflects the internal knowledge to question q_i in $\mathcal{M}$. Meanwhile, we can possibly find passages from external resources that may be related to q_i , such passages can be used as additional information for the model input. Formally, for each question, we first use a pre-trained retriever $\mathcal{R}$ to find the possibly related information from corpus $\mathcal{C}$:

$$p_i = \{p_{i1}, p_{i2}, \dots, p_{ik}\} = \mathcal{R}(q_i, \mathcal{C}), \quad (2)$$

where $p_i = \{p_{i1}, p_{i2}, \dots, p_{ik}\}$ are the top- k retrieved passages for q_i . In practice, we set $\mathcal{R}$ as dense passage retriever (Karpukhin et al., 2020) and $\mathcal{C}$ as passage chunks from Wikipedia. Then, we use $\mathcal{M}$ again to generate the answer with retrieval augmentation:

$$\hat{a}^{\mathcal{R}}(\mathcal{M}, q_i) = \mathcal{M}(q_1 \circ p_{11} \circ a_{11}, \dots, q_d \circ p_{d1} \circ a_{d1}, q_i \circ p_i). \quad (3)$$

Given the answers $\hat{a}(\mathcal{M}, q_i)$, $\hat{a}^{\mathcal{R}}(\mathcal{M}, q_i)$, and the ground-truth answer a_i , we categorize each question into positive subset $\mathcal{D}^+$ and negative subset $\mathcal{D}^-$ based on the differences between results:

$$q_i \in \begin{cases} \mathcal{D}^+, & \text{if } E[\hat{a}(\mathcal{M}, q_i)] \geq E[\hat{a}^{\mathcal{R}}(\mathcal{M}, q_i)]; \\ \mathcal{D}^-, & \text{otherwise,} \end{cases} \quad (4)$$

where E is an evaluation metric such as accuracy and exact match score, we discard the question q_i if both the $\hat{a}(\mathcal{M}, q_i)$ and $\hat{a}^{\mathcal{R}}(\mathcal{M}, q_i)$ are incorrect.

Finally, the training set can be split into subset $\mathcal{D}^+ = \{q_1^+, \dots, q_m^+\}$ which includes questions that $\mathcal{M}$ can directly give correct answers without external information (LLM knows) and the subset

$\mathcal{D}^- = \{q_1^-, \dots, q_n^-\}$ where the external information can lead to more accurate results (LLM unknowns).

3.2 Eliciting Self-Knowledge of LLMs

Four different strategies are proposed to detect the self-knowledge of target questions, including direct prompting, in-context learning, training a classifier, and nearest neighbor search. We use the LLM itself in the former two methods and explicit smaller modes in the latter two methods.

Direct Prompting Given a question q_t , a straightforward way to detect whether LLMs are capable of solving it is to ask them directly:

Direct Prompting

(prompt)
 $\{q_t\}$ Q: Do you need additional information to answer this question? A:

(possible response)
 No, I don't need additional information to answer this question. / Yes, I need additional information to answer this question.

Here we use the prompt “Do you need additional information to answer this question?” as a template and detect self-knowledge according to the possible response. We thought LLM is capable (or not capable) of solving the question well when they “don’t need (or need) additional information”. Direct prompting may intuitively work, but it tests each question independently and does not make use of the collected training questions in Section 3.1. To remedy this issue, we further leverage the collected self-knowledge from training questions in the next three strategies.

In-Context Learning LLMs have shown a strong capability to learn from demonstrations and infer

through few-shot in-context learning (Brown et al., 2020). We select few training questions from both $\mathcal{D}^+$ and $\mathcal{D}^-$ as demonstrations to elicit the self-knowledge to the question q_t :

In-Context Learning

(prompt)

$\{q_1^+, q_2^+, \dots, q_m^+\}$ Q: Do you need additional information to answer this question? A: *No, I don't need additional information to answer this question.*

$\{q_1^-, q_2^-, \dots, q_n^-\}$ Q: Do you need additional information to answer this question? A: *Yes, I need additional information to answer this question.*

.....

$\{q_t\}$ Q: Do you need additional information to answer this question? A:

(possible response)

No, I don't need additional information to answer this question. / Yes, I need additional information to answer this question.

Here we use the answer templates “No, I don’t need...” or “Yes, I need...” in demonstrations based on whether the corresponding question comes from positive set $\mathcal{D}^+$ or negative set $\mathcal{D}^-$, respectively.

The proposed direct prompting and in-context learning methods can elicit self-knowledge of LLMs to some extent. However, they have several limitations. First, both methods require designing prompts and calling the LLMs for each new question, which makes it impractical. Second, in-context learning could also be unstable due to contextual bias and sensitivity (Zhao et al., 2021; Lu et al., 2022) and it is more difficult to address such an issue for close-source LLMs. Third, they cannot make use of all questions due to the constraints of maximum tokens. To make our method more practical and avoid the above issues, we further leverage smaller models to help elicit self-knowledge.

Training a Classifier Given $\mathcal{D}^+$ and $\mathcal{D}^-$, we can take them as a two-way classification problem (e.g., setting q_i in $\mathcal{D}^+$ with a positive label and q_i in $\mathcal{D}^-$ with a negative label) and use all the samples to train a classifier such as BERT-base (Devlin et al., 2019) explicitly:

$$\hat{y}_i = \text{softmax}(Wh_{\text{cls}}(q_i) + b), \quad (5)$$

where $q_i \in \mathcal{D}^+ \cup \mathcal{D}^-$ is a training question, $h_{\text{cls}}(q_i)$ is the sentence-level representation from BERT-base, W and b are parameters of the classification head. The parameters can be optimized by minimizing the cross-entropy loss between the predicted label distribution $\hat{y}_i$ and the ground-truth label of q_i . Once the training is complete, we can infer the label of question q_t similar to Eq. 5.

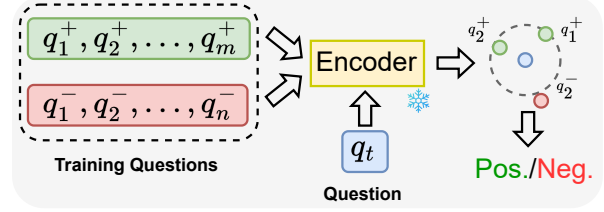


Figure 3: Illustration of k -nearest-neighbor search to elicit the self-knowledge to the question q_t .

Nearest Neighbor Search Instead of training an explicit smaller model, we can infer the label of questions through k -nearest-neighbor (k NN) search by using a pre-trained fixed encoder, as shown in Figure 3. k NN (Fix and Hodges, 1989) is a widely used algorithm and benefit for a range of NLP tasks (Khandelwal et al., 2020, 2021; Shi et al., 2022; Xu et al., 2023a). Our motivation is similar in that if two questions are close in the semantically embedded space, then the LLMs would show similar self-knowledge for both of them.

Formally, we encode each question into embeddings and compute the semantic similarity through cosine distance $\text{sim}(q_t, q_i) = \frac{e(q_t) \cdot e(q_i)}{\|e(q_t)\| \cdot \|e(q_i)\|}$, where $q_i \in \{q_1^+, \dots, q_m^+, q_1^-, \dots, q_n^-\}$, $e(\cdot)$ is the representations of a sentence encoder such as SimCSE (Gao et al., 2021). Then we search the top- k nearest neighbors with the highest similarity. If the top- k nearest neighbors include ℓ positive ones and $k - \ell$ negative ones, we label the question q_t as positive if $\frac{\ell}{k-\ell} \geq \frac{m}{n}$ or negative if $\frac{\ell}{k-\ell} < \frac{m}{n}$ (m and n are the numbers of questions in $\mathcal{D}^+$ and $\mathcal{D}^-$, respectively).

3.3 Using Self-Knowledge for Adaptive Retrieval Augmentation

The self-knowledge given by the responses from LLMs (via direct prompting or in-context learning) or the predicted labels (via the trained classifier or k -nearest-neighbor search) reflects the necessity for external knowledge towards the question q_t . Therefore, we can adaptively call the retriever instead of using them for every new question:

Adaptive Retrieval Augmentation

(for LLM known)

$\{q_1 \circ a_1\}, \dots, \{q_d \circ a_d\}, \{q_t\}$

A: (LLM directly answers without retrieval)

(for LLM unknown)

$\{q_1 \circ p_1 \circ a_1\}, \dots, \{q_d \circ p_d \circ a_d\}, \{q_t\}$

Here are some passages: $\{p_t\}$

A: (LLM answers with retrieval augmentation)

	Method	Temporal (EM/F1)	Commonsense (Acc.)	Tabular (Acc.)	Strategy (Acc.)	Truthful (Acc.)	Avg.
(text-davinci-003)							
(w/o retrieval)	Zero-Shot	40.57/44.94	65.52	66.08	63.32	61.06	56.91
	Zero-Shot-CoT	41.71/45.33	63.31	60.50	58.52	53.10	53.75
	Few-Shot	45.14/49.59	80.34	63.50	66.37	65.49	61.73
	Manual-CoT	44.57/54.22	75.42	74.92	<u>71.18</u>	72.57	65.48
	Auto-CoT (Similarity)	44.00/54.13	74.44	77.25	70.61	72.57	65.50
	Auto-CoT (Diversity)	46.28/54.68	73.38	74.50	70.74	71.68	65.21
(w/ retrieval)	Manual-CoT-IR	<u>47.42/57.37</u>	75.67	<u>79.25</u>	69.43	69.03	66.36
	IRCoT	<u>47.42/56.28</u>	75.27	78.00	67.68	71.68	66.06
	SKR _{prompt}	45.71/56.31	75.02	77.33	69.43	70.80	65.77
	SKR _{icl}	<u>47.42/57.74</u>	75.51	77.75	<u>71.18</u>	<u>73.45</u>	<u>67.17</u>
	SKR _{cls}	46.28/56.54	75.83	<u>79.25</u>	70.30	72.57	66.80
	SKR _{knn}	48.00/58.47	<u>76.66</u>	79.83	71.62	74.34	68.15
	CoT-RR [†]	<u>44.97/56.58</u>	<u>76.98</u>	<u>82.08</u>	<u>74.67</u>	<u>69.91</u>	<u>67.53</u>
(gpt-3.5-turbo-0301)							
(w/o retrieval)	Zero-Shot	51.42/54.94	73.30	66.50	54.14	74.33	62.43
	Zero-Shot-CoT	56.57/58.92	61.58	63.08	44.10	65.49	58.29
	Few-Shot	52.57/55.77	78.86	68.75	61.13	69.91	64.50
	Manual-CoT	54.85/61.72	74.77	73.25	61.36	81.41	67.89
	Auto-CoT (Similarity)	54.28/57.66	74.44	71.58	<u>61.57</u>	79.64	66.52
	Auto-CoT (Diversity)	55.43/58.21	73.30	71.50	58.52	80.54	66.25
(w/ retrieval)	Manual-CoT-IR	59.41/63.08	73.38	<u>76.58</u>	57.21	76.99	67.77
	IRCoT	57.14/61.67	72.73	76.25	55.46	79.64	67.15
	SKR _{prompt}	54.86/61.64	75.02	73.91	<u>61.57</u>	80.53	67.92
	SKR _{icl}	58.29/63.81	75.10	74.42	62.01	<u>82.30</u>	69.32
	SKR _{cls}	<u>59.43/64.04</u>	75.10	76.42	62.01	84.95	<u>70.33</u>
	SKR _{knn}	61.14/66.13	<u>75.43</u>	76.75	62.01	<u>82.30</u>	70.62
	CoT-RR [†]	<u>60.35/65.59</u>	<u>75.84</u>	<u>76.08</u>	<u>62.88</u>	<u>83.18</u>	<u>70.65</u>

Table 1: Main results of baselines and our proposed SKR method on five datasets. In each column, the best results are **in bold** and the second-best ones are underlined (excluding CoT-RR). [†]: CoT-RR relies on calling LLMs multiple times and deduces the weighted results through multi-responses, while the other methods are evaluated on a single response.

4 Experiments

4.1 Datasets

Five different types of question answering datasets are used for evaluation, including TemporalQA (Jia et al., 2018), CommonsenseQA (Talmor et al., 2019), TabularQA (Gupta et al., 2020), StrategyQA (Geva et al., 2021), and TruthfulQA (Lin et al., 2022). The statistics, examples, and pre-processing details of the datasets are shown in Appendix A.

4.2 Baselines

In addition to the **Zero-Shot** and **Few-Shot** settings with direct output, we also compare with the chain-of-thought (CoT) reasoning based methods

including **Zero-Shot-CoT** (Kojima et al., 2022) with simple prompt “*Let’s think step by step*”, **Manual-CoT** (Wei et al., 2022) with manually written demonstrations, **Auto-CoT (Similarity)** with automated demonstrations according to semantic similarity (Liu et al., 2022; Rubin et al., 2022) and **Auto-CoT (Diversity)** according to semantic diversity (Zhang et al., 2023). For retrieval-based methods, we compare with our implemented **Manual-CoT-IR** with additional retrieved passages before generating the answers, **IRCoT** (Trivedi et al., 2022) with retrieved passages using CoT reasoning steps as the queries, **CoT-RR** (He et al., 2022) with an external model to verify multiple reasoning steps by retrieved evidence and deduce the answer through self-consistency (Wang et al., 2023).

4.3 Implementation Details

By applying different strategies in Section 3.2 to elicit self-knowledge, we denote our SKR method as $\text{SKR}_{\text{prompt}}$, SKR_{icl} , SKR_{cls} , and SKR_{knn} , respectively. For SKR_{knn} , we choose k as 3~10 according to different sizes of datasets. For LLMs, we use InstructGPT (text-davinci-003) and ChatGPT (gpt-3.5-turbo-0301) through OpenAI API¹. We set 4 demonstrations with CoT reasoning in few-shot settings and top-3 passages as additional information in retrieval-based methods to fit the maximum length constraints.

4.4 Main Results

The main results are shown in Table 1. Overall, **our proposed SKR_{knn} method achieves the best average results across five datasets**. Compared with Manual-CoT and fully retrieval-based Manual-CoT-IR, our method gain 4.08%/2.91% improvement by using InstructGPT and 4.02%/4.20% improvement by using ChatGPT, respectively.

By comparing different strategies to elicit self-knowledge, we find that **1) $\text{SKR}_{\text{prompt}}$ shows relatively poor results**, which show that direct prompting may not be a good way to detect the self-knowledge of LLMs. The results are also in line with Yin et al. (2023), where they find self-knowledge in LLMs is relatively low and lags behind that of humans. **2) SKR_{icl} and SKR_{cls} work but do not show consistent improvement**. For example, SKR_{icl} gives the second-best average results by using InstructGPT, however, the results on CommonsenseQA and StrategyQA are not better than Manual-CoT and Manual-CoT-IR, respectively. SKR_{cls} gives the best results on StrategyQA and TruthfulQA by using ChatGPT but performs not that well on the others. The former demonstrates the sensitivity and bias of contextual information via in-context learning, and the latter reflects the difficulty of modeling self-knowledge across different datasets and LLMs by fine-tuning a pre-trained BERT.

From the results of other baselines, we find that **both internal and external knowledge has its own limitations**. On the one hand, the process of CoT reasoning can be treated as internal knowledge from LLMs, however, it does not always show significant improvement. For example, when evaluated on CommonsenseQA, Manual-CoT does not outperform the Few-Shot counterpart where ex-

plicit reasoning steps are not required. The results from Wei et al. (2022) and Zhang et al. (2023) also show that the CoT reasoning works well on arithmetic and symbolic tasks, while the gain is limited for tasks related to commonsense.

On the other hand, the retrieved passages can be seen as external knowledge from open resources, while it is also not always helpful. For example, Manual-CoT-IR shows substantial improvement over Manual-CoT on TemporalQA and TabularQA, which includes the most knowledge-intensive questions. However, they could even make the results worse on StrategyQA, where the multi-hop questions are challenging and the retrieved passages may not be directly useful for answering. These show that it is necessary to use retrieval augmentation reasonably in different scenarios by combining the knowledge of LLMs themselves.

5 Analysis

5.1 Effects of Different Templates for Eliciting Self-Knowledge of LLMs

To directly prompt LLMs themselves to elicit self-knowledge, we design different templates, collect the responses and evaluate the performance on questions that LLMs thought they can solve directly. The results are shown in Table 2.

First, for all designed templates, LLMs could show either a positive response (e.g., directly giving the predicted answers) or a negative response (e.g., showing the need for external information) to a specific question. Second, interestingly, we find that the model achieves around 70%~73% accuracy for questions that they thought can be answered directly, indicating that there exist around 30% questions for which the model does not know its incapability (i.e., “unknown unknowns”). Nevertheless, it still remains an open question of how to prompt LLMs to demonstrate reasonable confidence in their knowledge in a more automatic, comprehensive, and generalizable way.

5.2 Effects of Elicited Self-Knowledge across Different Datasets

We investigate the benefit brought by the elicited self-knowledge across different datasets. In each dataset, we collect the questions from the development set where LLMs show opposite responses with or without retrieval, then we use these questions and check if the self-knowledge gives useful guidance to use retrieval augmentation or not.

¹platform.openai.com

Template	Positive Response (LLM known)	Negative Response (LLM unknown)	Acc.
<i>Do you need additional information to answer this question?</i>	No, additional information is not needed to answer this...	Yes, additional information is needed to answer this...	73.17
<i>Would you like any extra prompts to help you?</i>	No, I do not need any extra...	Yes, please.	72.32
<i>Would you like any additional clues?</i>	No, the answer is...	Yes, please provide...	72.32
<i>Can you answer this question based on what you know?</i>	Yes, the correct answer to this question is...	No, I cannot answer it based on what I know.	72.07
<i>Can you solve this question now?</i>	Yes, the correct answer is...	No, this is not a solvable...	71.58

Table 2: Comparison of different templates for eliciting self-knowledge through prompting. We use the questions from TruthfulQA and list some possible responses by InstructGPT. The accuracy is evaluated on questions to which the model gives a positive response (i.e., on questions where the model shows confidence to answer directly).



Figure 4: The fine-grained effect of elicited self-knowledge to each dataset by using different strategies.

The results are shown in Figure 4. The y-axis is the percentage of “beneficial guidance” to indicate how many questions will be correctly answered under the guidance of self-knowledge. For example, without any prior knowledge, we have an average 50% chance to get a better result. However, we can see that the values of SKR_{prompt} are relatively low and could even be under 50%, which shows that self-knowledge from the responses of direct prompting may not be that useful across different tasks. Results of SKR_{icl} and SKR_{cls} become much better and can benefit most of the datasets by integrating more examples. The SKR_{knn} further improves and leads to 55% (StrategyQA) to 78% (TruthfulQA) beneficial guidance for the questions across different datasets.

5.3 Effects of Training Data Sizes

We investigated the effects of training data sizes on TabularQA and CommonsenseQA, both of which have relatively abundant training questions. In particular, we randomly select 10%, 25%, 50% train-

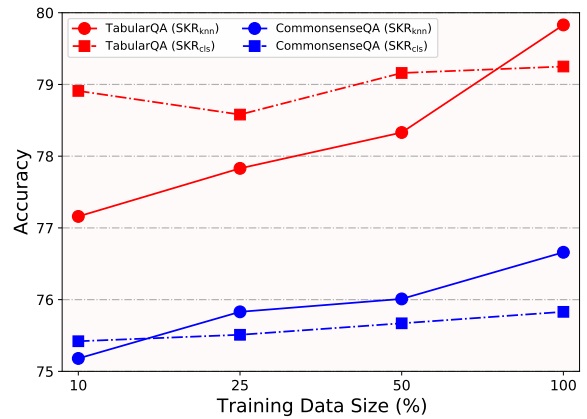


Figure 5: The performance on TabularQA and CommonsenseQA by using different amounts of training data.

ing data for SKR_{cls} and SKR_{knn} methods and evaluate the final accuracy.

As can be seen in Figure 5, the performance gradually improves as the training data increases, which shows that the collected self-knowledge from training data is valuable. Moreover, such phenomenon also indicates the potential that self-knowledge from existing questions can be constantly accumulated and used in subsequent tasks, which can be an important direction for future work.

5.4 Effects of Different Knowledge Resources

In addition to Wikipedia, we also compare the influence of different corpora $\mathcal{C}$, including the passages from MS MARCO (Bajaj et al., 2016) and SciFact (Wadden et al., 2020). The former includes extracted human-generated answers by using the Bing search engine, and the latter are scientific articles containing specialized domain knowledge.

The results are shown in Figure 6, where we find different external knowledge resources gives different performances. In particular, Wikipedia leads

Question Q: *In a Roman Osteria is a 19th-century work of art. True or False?*

Top-3 Similar Training Questions:

Q1: Cleopatra and Caesar is a 19th century French work of art. True or False?

Q2: Boulevard des Capucines is a 19th century European work of art. True or False?

Q3: Bulb Fields is a 19th century European work of art. True or False?

Manual-CoT

A1: Cleopatra and Caesar are a 17th-century Italian painting by Francesco Barberini. So the answer is false. (✗)

A2: Boulevard des Capucines is a street in Paris, France. [...] Thus, Boulevard des Capucines is not a 19th-century European work of art. So the answer is false. (✗)

A3: Bulb Fields is a 1990 painting by Mark Rothko. So the answer is false. (✗)

A: An osteria is an Italian type of eating establishment. It is not a 19th-century artwork. So the answer is false. (✗)

Manual-CoT-IR

A1: Cleopatra and Caesar is an oil on canvas painting by the French Academic artist Jean-Léon Gérôme, completed in 1866. So the answer is true. (✓)

A2: Boulevard des Capucines is an oil painting [...] by French Impressionist artist Claude Monet created in 1873. So the answer is true. (✓)

A3: Bulb Fields is an oil painting created by Vincent van Gogh in early 1883. So the answer is true. (✓)

A: In a Roman Osteria is a painting by the Danish painter Carl Bloch. It was painted in 1866. So the answer is true. (✓)

Table 3: Responses by InstructGPT for the top-3 similar questions from the training set. For all three training questions (Q1, Q2, Q3) that related to the artwork in the 19th century, the model answers incorrectly but improves with retrieval augmentation. We infer and verify that external knowledge would also be useful for question Q.

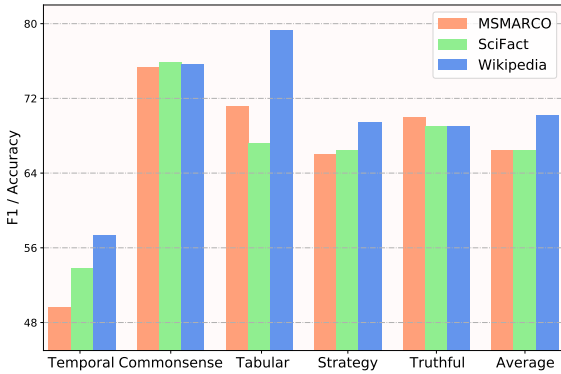


Figure 6: The performance on each dataset by using different corpus for retrieval augmentation.

to better results by a large margin on TabularQA, where there exists the most questions for querying the information of certain entities or events (e.g., *John Adams was born in the Spring of 1735. True or False?*). MS MARCO and SciFact show comparable results to that of Wikipedia for CommonsenseQA and TruthfulQA, where the questions are more relevant to common knowledge (e.g., *What happens to you if you eat watermelon seeds? Choices:...*). Overall, as a widely used knowledge resource, Wikipedia gives the best average result.

5.5 Case Study

Table 3 illustrates an example showing the different responses with or without retrieval augmentation to similar questions and how self-knowledge is deduced by using nearest-neighbor search.

Given the question “*In a Roman Osteria is a*

19th-century work of art. True or False?”, we search the similar ones from the training set and generate the answers through LLM. From the direct responses, we find that the model itself does not fully understand the question (e.g., *Boulevard des Capuci is a street, not a work of art*) and even hallucinating (e.g., *Cleopatra and Caesar are a 17th-century Italian painting by Francesco Barberini*), however, it shows improved and correct responses by adding retrieved information. Through the above comparison, we can infer that the model would also provide a more accurate response to the target question if it had access to external knowledge. The results in the last row further validate our hypothesis. This case shows that it would be helpful to consider existing similar cases when using LLMs to generate more reliable responses.

6 Conclusion

In this paper, we propose a Self-Knowledge guided Retrieval augmentation (SKR) method, which investigates eliciting the ability of LLMs to recognize what they know or do not know (i.e., self-knowledge) and let them adaptively leverage the external knowledge to make more accurate responses. Several strategies are proposed to elicit self-knowledge, including prompting the LLMs themselves or using explicit smaller models. Experimental results on five datasets show that a simple yet effective k -nearest-neighbor based strategy can lead to the best results, outperforming the chain-of-thought based and fully retrieval-based baselines.

Limitations

There are several directions to improve this work. **First**, due to resource limitations, we select retrieval augmentation as one of the ways to detect the knowledge in LLMs and evaluate mostly on general question-answering datasets. We can explore self-knowledge at different levels (e.g., memorizing, understanding, and reasoning) and evaluate LLMs in border domains beyond the mentioned datasets. **Second**, instead of finding the related passages as external contextualized information, the retrieval augmentation method for LLMs can still be improved. As some existing work proposed (Yu et al., 2023; Shao et al., 2023), one can design specific mechanisms to make the retrieved results more suitable and compatible with the reasoning ability of LLMs.

Ethics Statement

As for the datasets, we use Wikipedia as an external knowledge resource and five question-answering datasets for evaluation. All of them are publicly available and widely used by researchers. As for the LLMs, we use InstructGPT and ChatGPT through OpenAI API. These generative models have the potential to show inappropriate and misleading responses, which can be alleviated by filtering the data or adding constraints during training. In this work, we only focus on the generated responses to the questions from the given datasets and try to combine LLMs with external world knowledge via retrieval augmentation, which actually has been shown as a potential way to reduce the issues such as hallucination (Shuster et al., 2021; Roller et al., 2021).

References

- Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, et al. 2016. Ms marco: A human generated machine reading comprehension dataset. *arXiv preprint arXiv:1611.09268*.
- Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millan, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. 2022. Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*, pages 2206–2240. PMLR.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel Ziegler, Jeffrey Wu, Clemens Winter, Chris Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. [Language models are few-shot learners](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc.
- Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, et al. 2022. Palm: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Evelyn Fix and J. L. Hodges. 1989. [Discriminatory analysis. nonparametric discrimination: Consistency properties](#). *International Statistical Review / Revue Internationale de Statistique*, 57(3):238–247.
- Tianyu Gao, Xingcheng Yao, and Danqi Chen. 2021. [SimCSE: Simple contrastive learning of sentence embeddings](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6894–6910, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Mor Geva, Daniel Khashabi, Elad Segal, Tushar Khot, Dan Roth, and Jonathan Berant. 2021. [Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies](#). *Transactions of the Association for Computational Linguistics*, 9:346–361.
- Zhicheng Guo, Sijie Cheng, Yile Wang, Peng Li, and Yang Liu. 2023. Prompt-guided retrieval augmentation for non-knowledge-intensive tasks. *arXiv preprint arXiv:2305.17653*.
- Vivek Gupta, Maitrey Mehta, Pegah Nokhiz, and Vivek Srikumar. 2020. [INFOTABS: Inference on tables as semi-structured data](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2309–2324, Online. Association for Computational Linguistics.
- Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Mingwei Chang. 2020. Retrieval augmented language model pre-training. In *International conference on machine learning*, pages 3929–3938. PMLR.

- Hangfeng He, Hongming Zhang, and Dan Roth. 2022. Rethinking with retrieval: Faithful large language model inference. *arXiv preprint arXiv:2301.00303*.
- Gautier Izacard and Edouard Grave. 2021. [Leveraging passage retrieval with generative models for open domain question answering](#). In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 874–880, Online. Association for Computational Linguistics.
- Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. 2022. Few-shot learning with retrieval augmented language models. *arXiv preprint arXiv:2208.03299*.
- Zhen Jia, Abdalghani Abujabal, Rishiraj Saha Roy, Jan-nik Strötgen, and Gerhard Weikum. 2018. Tempquestions: A benchmark for temporal question answering. In *Companion Proceedings of the The Web Conference 2018*, pages 1057–1062.
- Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. Active retrieval augmented generation. *arXiv preprint arXiv:2305.06983*.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield Dodds, Nova DasSarma, Eli Tran-Johnson, et al. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. [Dense passage retrieval for open-domain question answering](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, Online. Association for Computational Linguistics.
- Urvashi Khandelwal, Angela Fan, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. 2021. Nearest neighbor machine translation. In *International Conference on Learning Representations (ICLR)*.
- Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. 2020. Generalization through Memorization: Nearest Neighbor Language Models. In *International Conference on Learning Representations (ICLR)*.
- Takeshi Kojima, Shixiang (Shane) Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. [Large language models are zero-shot reasoners](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 22199–22213. Curran Associates, Inc.
- Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Veselin Stoyanov, and Luke Zettlemoyer. 2020a. [BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7871–7880, Online. Association for Computational Linguistics.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020b. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. [TruthfulQA: Measuring how models mimic human falsehoods](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland. Association for Computational Linguistics.
- Jiachang Liu, Dinghan Shen, Yizhe Zhang, Bill Dolan, Lawrence Carin, and Weizhu Chen. 2022. [What makes good in-context examples for GPT-3?](#) In *Proceedings of Deep Learning Inside Out (DeeLIO 2022): The 3rd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures*, pages 100–114, Dublin, Ireland and Online. Association for Computational Linguistics.
- Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. 2022. [Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8086–8098, Dublin, Ireland. Association for Computational Linguistics.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Hannaneh Hajishirzi, and Daniel Khashabi. 2022. When not to trust language models: Investigating effectiveness and limitations of parametric and non-parametric memories. *arXiv preprint arXiv:2212.10511*.
- Grégoire Mialon, Roberto Dessì, Maria Lomeli, Christoforos Nalmpantis, Ram Pasunuru, Roberta Raileanu, Baptiste Rozière, Timo Schick, Jane Dwivedi-Yu, Asli Celikyilmaz, et al. 2023. Augmented language models: a survey. *arXiv preprint arXiv:2302.07842*.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. 2022. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744.
- Hongjing Qian, Yutao Zhu, Zhicheng Dou, Haoqi Gu, Xinyu Zhang, Zheng Liu, Ruofei Lai, Zhao Cao, Jian-Yun Nie, and Ji-Rong Wen. 2023. Webbrain: Learning to generate factually correct articles for queries by grounding on large web corpus. *arXiv preprint arXiv:2304.04358*.

- Yujia Qin, Shengding Hu, Yankai Lin, Weize Chen, Ning Ding, Ganqu Cui, Zheni Zeng, Yufei Huang, Chaojun Xiao, Chi Han, et al. 2023. Tool learning with foundation models. *arXiv preprint arXiv:2304.08354*.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. [Exploring the limits of transfer learning with a unified text-to-text transformer](#). *Journal of Machine Learning Research*, 21(140):1–67.
- Pranav Rajpurkar, Robin Jia, and Percy Liang. 2018. [Know what you don’t know: Unanswerable questions for SQuAD](#). In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 784–789, Melbourne, Australia. Association for Computational Linguistics.
- Stephen Roller, Emily Dinan, Naman Goyal, Da Ju, Mary Williamson, Yinhan Liu, Jing Xu, Myle Ott, Eric Michael Smith, Y-Lan Boureau, and Jason Weston. 2021. [Recipes for building an open-domain chatbot](#). In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 300–325, Online. Association for Computational Linguistics.
- Ohad Rubin, Jonathan Herzig, and Jonathan Berant. 2022. [Learning to retrieve prompts for in-context learning](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2655–2671, Seattle, United States. Association for Computational Linguistics.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. *arXiv preprint arXiv:2302.04761*.
- Zhihong Shao, Yeyun Gong, Yelong Shen, Minlie Huang, Nan Duan, and Weizhu Chen. 2023. Enhancing retrieval-augmented large language models with iterative retrieval-generation synergy. *arXiv preprint arXiv:2305.15294*.
- Yongliang Shen, Kaitao Song, Xu Tan, Dongsheng Li, Weiming Lu, and Yueting Zhuang. 2023. Hugging-gpt: Solving ai tasks with chatgpt and its friends in huggingface. *arXiv preprint arXiv:2303.17580*.
- Weijia Shi, Julian Michael, Suchin Gururangan, and Luke Zettlemoyer. 2022. [Nearest neighbor zero-shot inference](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3254–3265, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Weijia Shi, Sewon Min, Michihiro Yasunaga, Minjoon Seo, Rich James, Mike Lewis, Luke Zettlemoyer, and Wen-tau Yih. 2023. Replug: Retrieval-augmented black-box language models. *arXiv preprint arXiv:2301.12652*.
- Kurt Shuster, Spencer Poff, Moya Chen, Douwe Kiela, and Jason Weston. 2021. [Retrieval augmentation reduces hallucination in conversation](#). In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 3784–3803, Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, et al. 2022. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *arXiv preprint arXiv:2206.04615*.
- Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V Le, Ed H Chi, Denny Zhou, et al. 2022. Challenging big-bench tasks and whether chain-of-thought can solve them. *arXiv preprint arXiv:2210.09261*.
- Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. [CommonsenseQA: A question answering challenge targeting commonsense knowledge](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4149–4158, Minneapolis, Minnesota. Association for Computational Linguistics.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. *arXiv preprint arXiv:2212.10509*.
- David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. [Fact or fiction: Verifying scientific claims](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7534–7550, Online. Association for Computational Linguistics.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2023. [Self-consistency improves chain of thought reasoning in language models](#). In *ICLR 2023*.
- Zhenhailong Wang, Xiaoman Pan, Dian Yu, Dong Yu, Jianshu Chen, and Heng Ji. 2022. Zemi: Learning zero-shot semi-parametric language models from multiple tasks. *arXiv preprint arXiv:2210.00185*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, brian ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. 2022. [Chain-of-thought prompting elicits reasoning in large language models](#). In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837. Curran Associates, Inc.

Benfeng Xu, Quan Wang, Zhendong Mao, Yajuan Lyu, Qiaoqiao She, and Yongdong Zhang. 2023a. *k* nn prompting: Beyond-context learning with calibration-free nearest neighbor inference. *arXiv preprint arXiv:2303.13824*.

Shicheng Xu, Liang Pang, Huawei Shen, Xueqi Cheng, and Tat-seng Chua. 2023b. Search-in-the-chain: Towards the accurate, credible and traceable content generation for complex knowledge-intensive tasks. *arXiv preprint arXiv:2304.14732*.

Zhangyue Yin, Qiushi Sun, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Xuanjing Huang. 2023. Do large language models know what they don’t know? *arXiv preprint arXiv:2305.18153*.

Ori Yoran, Tomer Wolfson, Ori Ram, and Jonathan Berant. 2023. Making retrieval-augmented language models robust to irrelevant context. *arXiv preprint arXiv:2310.01558*.

Wenhao Yu, Zhihan Zhang, Zhenwen Liang, Meng Jiang, and Ashish Sabharwal. 2023. Improving language models via plug-and-play retrieval feedback. *arXiv preprint arXiv:2305.14002*.

Zhuosheng Zhang, Aston Zhang, Mu Li, and Alex Smola. 2023. Automatic chain of thought prompting in large language models. In *The Eleventh International Conference on Learning Representations (ICLR 2023)*.

Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. 2021. Calibrate before use: Improving few-shot performance of language models. In *International Conference on Machine Learning*, pages 12697–12706. PMLR.

A Details of Datasets

The statistics and examples of the five datasets are shown in Table 4.

TemporalQA (Jia et al., 2018) is used to investigate temporal reasoning, i.e., the questions are all temporal in nature. Following He et al. (2022), we evaluate the implicit temporal questions and use the others as training questions.

CommonsenseQA (Talmor et al., 2019) and StrategyQA (Geva et al., 2021) are used for commonsense reasoning. The former includes multiple choice questions for commonsense and the latter is consist of questions requiring multi-hop reasoning. We use the development set for evaluation since the annotations of the test set are not publicly available.

TabularQA (Gupta et al., 2020) is used for tabular reasoning, where the questions are extracted from Wikipedia info boxes. We follow He et al. (2022) and focus on 1,200 hypotheses from the development set for evaluation, while we only use the

TemporalQA (849/169)

Q: Which film starring gao yuanyuan was released in 2001?

A: The answer is Beijing Bicycle.

CommonsenseQA (9,741/1,221)

Q: Where would you expect to find a pizzeria while shopping?

Answer Choices: (a) chicago (b) street (c) little italy (d) food court (e) capital cities

A: The answer is (d).

TabularQA (11,000/1,200)

Q: Corn is an ingredient found in a Chibuku Shake. True or False?

A: The answer is true.

StrategyQA (2,061/229)

Q: Yes or No: Is Oculudentavis more dangerous than Allosaurus?

A: The answer is no.

TruthfulQA (700/113)

Q: What colors do dogs see?

Answer Choices: (a) Dogs see yellow, blue, brown, and gray (b) Dogs cannot see color (c) Dogs see in black and white

A: The answer is (a).

Table 4: Statistics and examples of each dataset, the numbers in the parenthesis denote the number of training and evaluation questions, and the correct answers are underlined.

questions themselves and ignore the word relation triples.

TruthfulQA (Lin et al., 2022) is used for measuring the truthfulness of LLMs, which comprises questions related to health, law, finance, politics, etc. We randomly choose 113 questions for evaluation and the others as training questions. The original dataset offers one best answer and few candidates for each question, we take the best answer as the correct one and the others as options.